

# Challenge Write-Up Template

## Challenge Information

|                     |                         |
|---------------------|-------------------------|
| Student Name        | Evallo, Daniella Kim N. |
| Section / Class     | BSIT-3G                 |
| Challenge Name      | Printer Shares          |
| Category            | General Skills          |
| Points / Difficulty | Easy                    |
| Date Solved         | 08/09/2026              |
| Flag Format Given   | picoCTF{...}            |

## 1. Challenge Description

The challenge asks me to retrieve an important file that was accidentally sent to a network printer. The printer service is exposed on port 52274, and the goal is to access the print server, locate the shared file, and recover the flag.

## 2. Initial Analysis / Reconnaissance

I first reviewed the challenge page and noted the printer port (52274) and the provided netcat connectivity test. The challenge is a General Skills task with an Easy difficulty. The hints point toward the SMB protocol and specifically suggest using smbclient or smbutil, so I focused on enumerating SMB shares and retrieving the file from the accessible share.

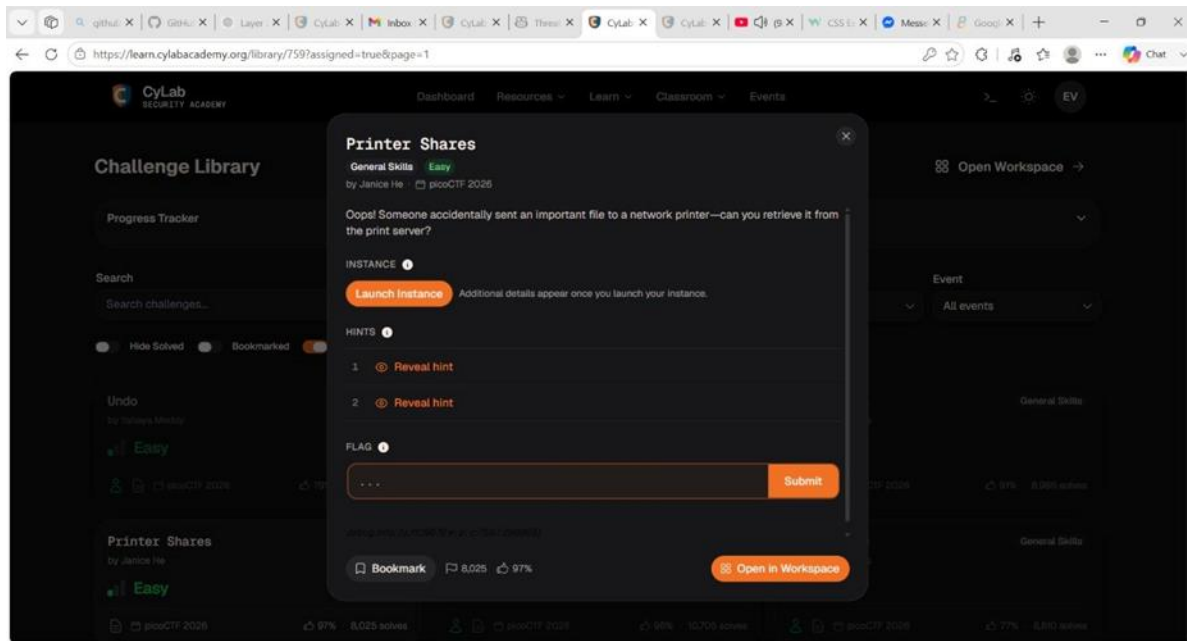
## 3. Tools and Commands Used

| Tool / Command                                            | What it was used for                                   | Result / Output                                                                       |
|-----------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------|
| nc -vz mysterious-sea.picoctf.net 52274                   | Check whether the printer service is reachable.        | Connection to mysterious-sea.picoctf.net (3.130.79.223) 52274 port [tcp/*] succeeded! |
| smbclient -L //mysterious-sea.picoctf.net -p 52274 -N     | Enumerate the available SMB shares without a password. | The server exposed a share named shares, described as Public Share With Guests.       |
| smbclient //mysterious-sea.picoctf.net/shares -p 52274 -N | Connect to the public SMB share.                       | Connected to the shares SMB share and received an smb prompt.                         |
| ls                                                        |                                                        | List files available inside the SMB share.                                            |
| get flag.txt                                              | Download flag.txt from the SMB share                   | flag.txt was retrieved successfully.                                                  |
| cat flag.txt                                              | Read the downloaded flag file.                         | The flag was displayed in the terminal.                                               |

## 4. Solution Walkthrough

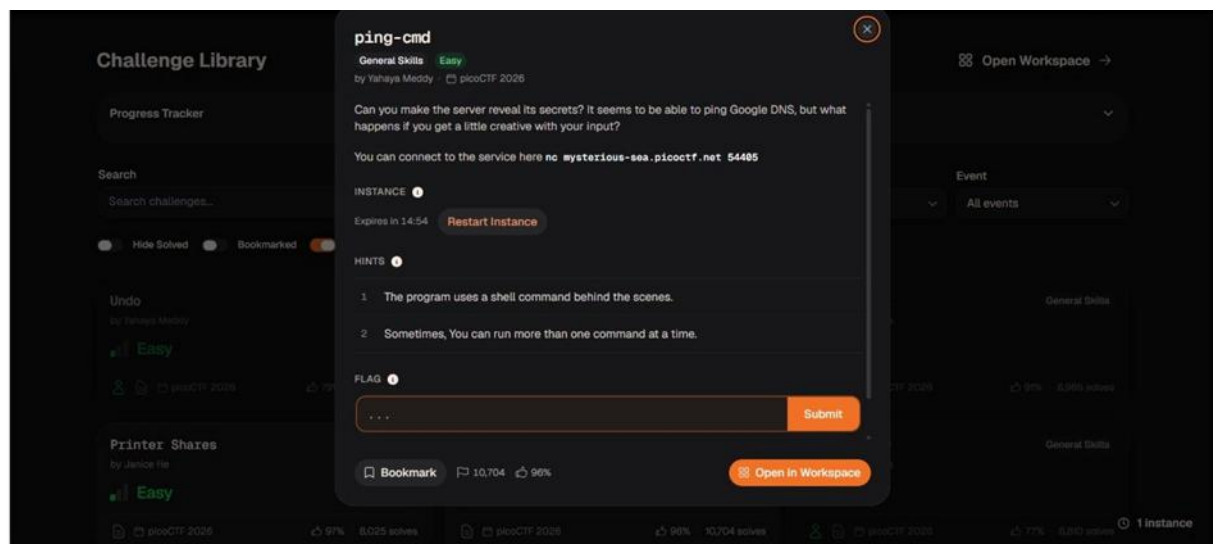
### Step 1

I began from the challenge page and launched the instance. Once the instance was running, the page showed the printer service on port 52274 and the option to restart the instance. Starting from a fresh instance ensures the environment is in the expected state.



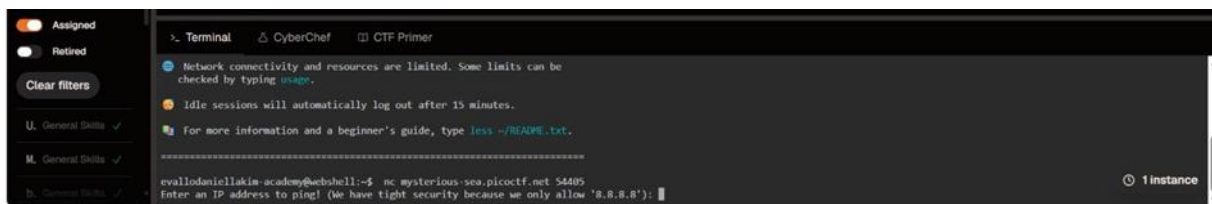
### Step 2

The challenge explains that the server can ping Google DNS and asks what happens with a more creative input. The hints are important: the program uses a shell command behind the scenes, and sometimes more than one command can be run at a time.



### Step 3

I connected to the provided service with netcat using **nc mysterious-sea.picoctf.net 54405**. The terminal then displayed the prompt asking for an IP address and stated that only **8.8.8.8** was allowed.



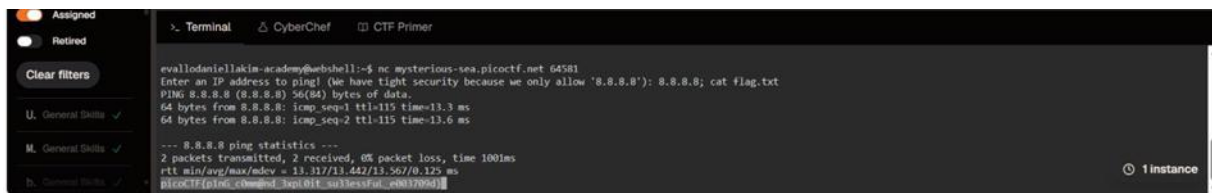
## Step 4

Instead of entering only the allowed IP address, I appended a second command using a semicolon: **8.8.8.8; ls**. The ping still ran, and the output after it showed **flag.txt** and **script.sh**. This confirmed that the input was being interpreted by a shell and that an additional command could be executed.



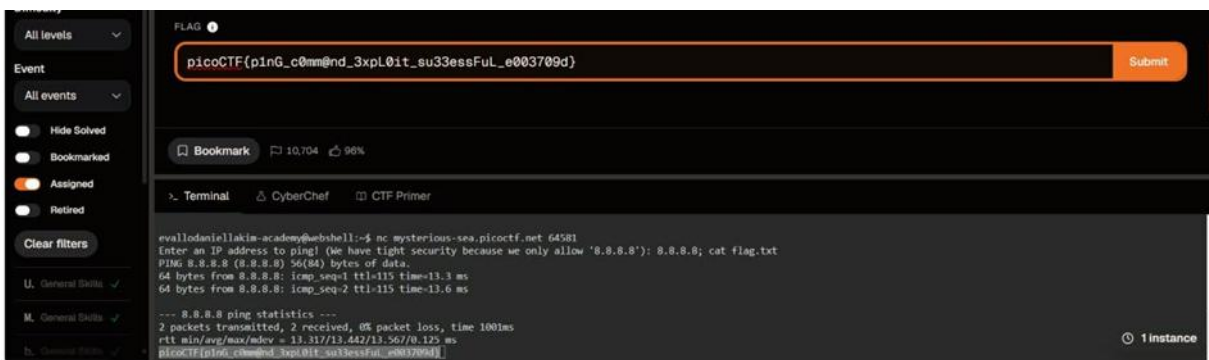
## Step 5

With flag.txt now visible, used the shell behavior to read the file with **cat flag.txt**. The terminal displayed the recovered picoCTF flag.



## Step 6

I copied the recovered flag exactly and entered it into the challenge submission field. The final screenshot shows the flag present in the submission box.



## 5. Flag

picoCTF{p1nG\_c0mm@nd\_3xpL0it\_su3essFuL\_e003709d}

## 6. Key Concept / What I Learned

I learned how shell command injection can occur when user input is passed into a shell command without proper validation. In this challenge, adding a second command after the allowed IP address caused the service to execute an additional command and expose files that were not meant to be directly listed.

## 7. References

[picoCTF 2026 | Denton's Blog](#)

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